ONCOSENTINEL - PRODUCT REQUIREMENTS DOCUMENT

PRD v2.0 - Research-gap aligned

Version 2.0 | Research-gap aligned baseline | Body text 10 pt

1. Product Overview

OncoSentinel v2.0 is a multimodal cancer AI and clinical decision-support platform that combines specialized disease models, intelligent routing, evidence-grounded retrieval, a local reasoning layer, federated learning, clinically grounded explainability, longitudinal patient context, uncertainty-aware safety controls, fairness evaluation, and clinical outcome measurement.

2. Product Vision

Build a trustworthy oncology intelligence platform that does not merely predict a disease, but determines when a prediction is appropriate, quantifies uncertainty, explains clinically meaningful evidence, incorporates authorized longitudinal context, detects subgroup disparities, supports human review, and measures whether the product improves real clinical work.

3. Research Problem

The supplied literature shows a persistent gap between predictive capability and clinical translation. Earlier cancer ML reviews identify limited external validation, small samples, and weak clinical adoption [Kourou et al., 2015]. Federated learning improves access to diverse data and out-of-sample generalization [Pati et al., 2022]. Explainability research shows that quantitative explanation metrics are needed, while clinically meaningful concept-based explanations remain underdeveloped [Sangwan, 2024]. End-to-end clinical phenotyping demonstrates the value of workflow alignment but still requires prospective clinical-impact evaluation [Shyr et al., 2026]. OncoSentinel v2.0 is positioned to address these gaps as a product and research platform.

4. Product Goals

1. Provide safe multimodal cancer analysis across the authoritative eight-category taxonomy.
2. Prevent irrelevant or unsupported inputs from reaching inappropriate ML models.
3. Use federated learning to support multi-institutional model development without centralized raw patient-data movement.
4. Make uncertainty and abstention first-class product behaviors.
5. Generate clinically grounded, concept-based explanations tied to evidence and counter-evidence.
6. Use longitudinal patient context when authorized and disclose the historical evidence used.
7. Continuously evaluate subgroup fairness and route-level disparities.
8. Measure product value through clinical and workflow outcomes rather than accuracy alone.
9. Maintain complete provenance, versioning, auditability and safe-failure behavior.

5. Target Users

User	Primary value
Clinician	Review multimodal evidence, prediction, uncertainty, explanation, longitudinal context and structured reports.
Researcher	Run experiments, inspect datasets/models, compare sites, analyze fairness/uncertainty and outcomes.
Clinical study reviewer	Review endpoint definitions, adjudication, safety signals and subgroup results.

User	Primary value
Administrator	Manage access, models, datasets, governance, deployment and audit.
Analyst	Monitor operational, model, fairness, drift and outcome dashboards.
Auditor	Inspect provenance, decisions, evidence, model versions and safety trails.

6. Product Principles

- Evidence before generation.
- Routing before execution.
- Uncertainty before autonomy.
- Clinical concepts before opaque explanations.
- Longitudinal context before isolated snapshots when authorized.
- Fairness must be measured, not assumed.
- Clinical impact must be measured, not inferred from model accuracy.
- Safe failure is a product feature.
- Version everything: data, models, prompts, calibration, explanation policies, thresholds and evaluation studies.

7. Core User Journeys

7.1 Multimodal image analysis

Upload image/PDF -> validate -> modality/quality detection -> relevance gate -> disease category -> model head -> calibrated prediction -> uncertainty/OOD -> longitudinal context -> concept-based explanation -> evidence verification -> clinician review -> report -> outcome capture.

7.2 Document knowledge analysis

Upload PDF -> validate -> extract text/pages/images -> intent classification -> RAG -> vector search -> rerank -> evidence check -> local LLM -> citation/provenance -> verifier -> answer/report.

7.3 Longitudinal case review

Open case -> render timeline -> select relevant episodes -> model/reasoning uses authorized history -> show timepoints influencing result -> clinician review -> update timeline/outcome.

7.4 Uncertain result

Prediction -> uncertainty/OOD -> if above threshold: review or abstain -> present reason for uncertainty -> request additional evidence when possible -> no forced diagnosis.

7.5 Fairness dashboard

Model release -> subgroup evaluation -> compare performance/calibration/error/abstention -> review disparities -> mitigation decision -> approval gate or hold release.

7.6 Clinical outcome study

Prospective study enrollment -> baseline workflow -> AI-assisted workflow -> timestamped decisions -> clinician actions -> endpoint collection -> adjudication -> subgroup analysis -> final study report.

8. Product Feature Set

Area	Features
Core AI	8 category heads, shared backbone, intelligent routing, multimodal ingestion, model registry

Area	Features
Safety	uncertainty quantification, calibration, OOD detection, abstention, safe-failure states
Clinical XAI	concept extraction, evidence linking, counter-evidence, technical attribution, explanation evaluation
Longitudinal	timeline, event provenance, trend features, treatment-response linkage, temporal integrity
Knowledge	Vector DB, knowledge graph, RAG, reranking, evidence scoring, citation grounding
Privacy/Federation	federated coordinator, secure aggregation, local training, privacy controls, node governance
Fairness	subgroup metrics, calibration gap analysis, error disparity reports, release gates
Clinical research	external validation, prospective pilots, outcome capture, comparative evaluation
Platform	RBAC, cases, reports, audit, monitoring, dashboards, CI/CD, deployment, rollback

9. Product Differentiation

The differentiator is not the number of models. It is the coordinated decision fabric that determines which capability should execute, how certain the result is, how the result can be explained clinically, how longitudinal context changes interpretation, whether performance is equitable, and whether the system improves the workflow.

10. Updated Product Roadmap

Phase	Product milestone	Major deliverables
1	Research-safe ML MVP	Breast + Kidney; multimodal ingestion; routing; calibration/uncertainty baseline; model registry; external-validation harness
2	Multi-Cancer Engine	All 8 categories; authoritative taxonomy; category heads; fairness evaluation skeleton; timeline data model
3	Clinical Intelligence	RAG/KG; local LLM; concept-based XAI; longitudinal review; evidence verifier; reports
4	Federated Trust Layer	Federated coordinator; secure aggregation; drift monitoring; continual-learning gates; fairness/calibration release controls
5	Clinical Evaluation	Prospective pilot; outcome capture; comparative workflow study; clinician usability/trust evaluation; hospital integration

11. MVP Definition v2.0

The first research-grade MVP is not only "prediction works". For a supported Breast or Kidney case, the platform must validate input quality, route correctly, provide a calibrated prediction, quantify uncertainty, generate a basic clinically grounded explanation, store provenance, and support safe clinician review.

12. Success Metrics

Metric group	Example metrics
Model	AUC, sensitivity, specificity, F1, Dice, calibration error
Uncertainty	selective risk, abstention precision, OOD detection AUROC where appropriate, calibration
Explainability	faithfulness, robustness, localization, clinician usefulness, agreement with clinical concepts
Longitudinal	event retrieval accuracy, timeline integrity, temporal leakage rate, trajectory prediction metrics
Fairness	subgroup sensitivity/specificity, calibration gap, FNR/FPR gap, abstention gap
Workflow	time-to-diagnosis, time-to-review, workload, completion rate, override rate
Clinical	decision quality, unsafe miss rate, appropriate escalation, study-specific downstream outcomes
Platform	availability, latency, error rate, retrieval precision/recall proxies, audit completeness

13. Product Risks and Mitigations

Risk	Mitigation
Overconfident wrong prediction	calibration + uncertainty + OOD + abstention + human review
Clinically misleading explanation	concept-based XAI + evidence linking + clinician evaluation
Historical context causing leakage	timestamp-aware datasets and temporal split controls
Subgroup performance disparity	mandatory fairness evaluation and release gate
Model drift	drift detection + scheduled and event-triggered evaluation + rollback
Clinical value not demonstrated	prospective outcome study and comparative workflow evaluation
Hallucinated knowledge response	evidence gate + grounded local LLM + verifier + NO_GROUNDED_ANSWER
Privacy exposure	federated/local processing, access controls, minimization, audit, approved privacy mechanisms

14. Definition of Done - Research-grade Product

- All eight categories are versioned and routable through the authoritative taxonomy.
- Supported multimodal input flows are validated and safe-fail paths are operational.
- Predictions expose uncertainty, calibration state and OOD status.

- Clinical explanations are concept-based, evidence-linked and quantitatively evaluated.
- Longitudinal timelines are immutable, provenance-linked and usable in authorized reasoning.
- Fairness evaluation is mandatory before production-oriented model promotion.
- Federated learning and multi-site external validation are operational at research scale.
- Clinical outcomes can be captured under a defined prospective study protocol.
- Monitoring includes performance, drift, fairness, uncertainty, safety and operational metrics.
- Every production-oriented model release has dataset/model/provenance/evaluation/rollback artifacts.

15. Updated Product Positioning

OncoSentinel should be positioned as a multimodal cancer AI and clinical decision-support research platform that combines specialized ML, evidence-grounded knowledge retrieval, controlled AI assistance, federated learning, uncertainty-aware safety, clinically grounded explainability, longitudinal reasoning, fairness evaluation, and clinical outcome measurement. Its promise is not autonomous medical authority; it is an auditable, evidence-grounded and uncertainty-aware decision-support fabric that can be scientifically evaluated.

16. Literature References

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